mocha+3.0.5(dicoms)

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MOCHA+ — Preprocess, data loading, 3D Tracing

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1. Preprocess

Both entries sit under the top menu **Preprocess**.

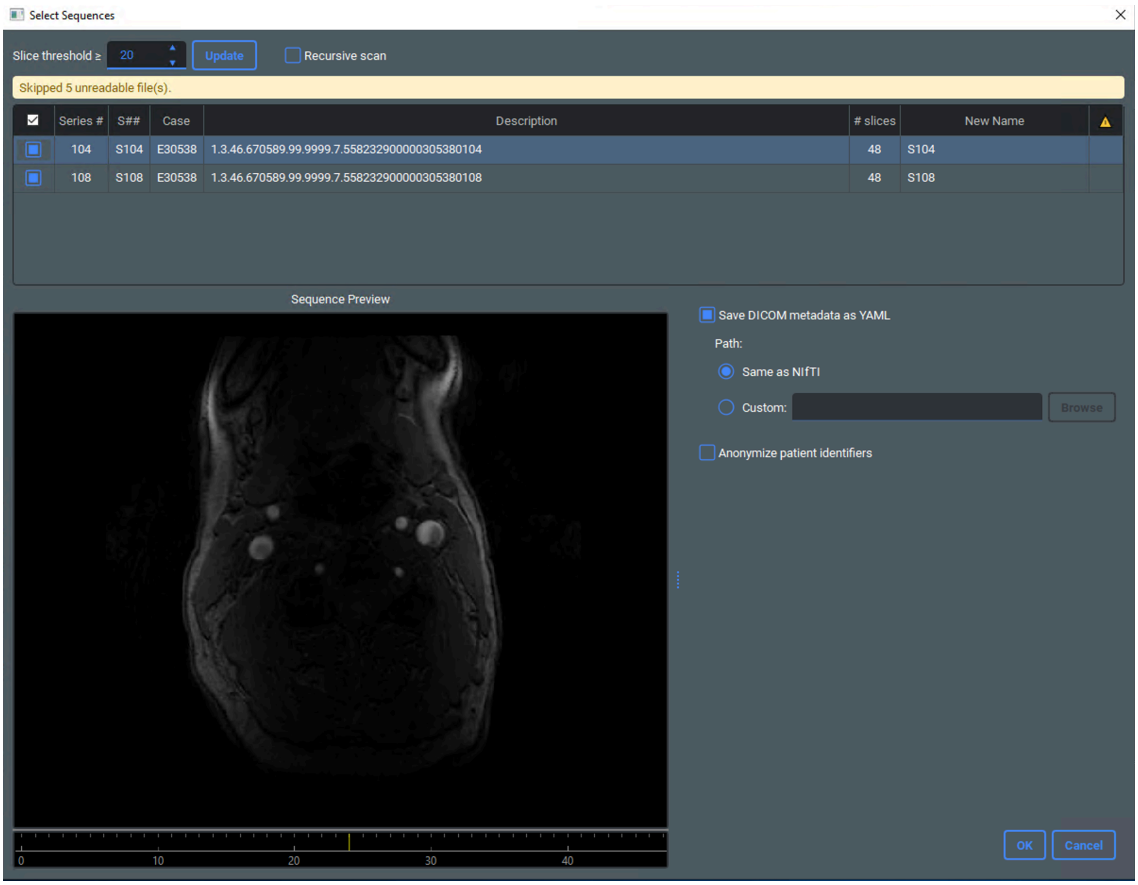
1.1 Preprocess → MV: Open Raw Data

Purpose: Batch-convert a **DICOM tree** to `*.nii.gz`, then optionally run the same registration chain as **Registration**. Does **not** write `.workflow_state.yaml` (no tracked workflow session).

Steps

1. Unless paths are fixed in code: **Select DICOM Directory** → pick the DICOM root.
2. **Select/Create Output Directory** → pick or create the folder where all `.nii.gz` files are written.

3. Dialog **Select Sequences** (`DialogSequenceSelection`):
    - Top row: **Slice threshold ≥**, **Update**, **Recursive scan** (recursive subdirectory scan when checked). **Update** re-runs the scan with the current threshold.
    - Table headers: **Series #**, **S##**, **Case**, **Description**, **# slices**, **New Name**, ⚠️; the first column selects whether each series is imported.
    - **Sequence Preview**; optional YAML metadata export (**Same as NIfTI** / **Custom)**



4. After confirmation: progress **Converting sequences...**; on success, the app continues into the registration flow in **section 1.2**.

Outcome: `{New Name}.nii.gz` per imported series in the output folder; registration dialogs follow unless the flow is cancelled earlier.

1.2 Preprocess → Registration

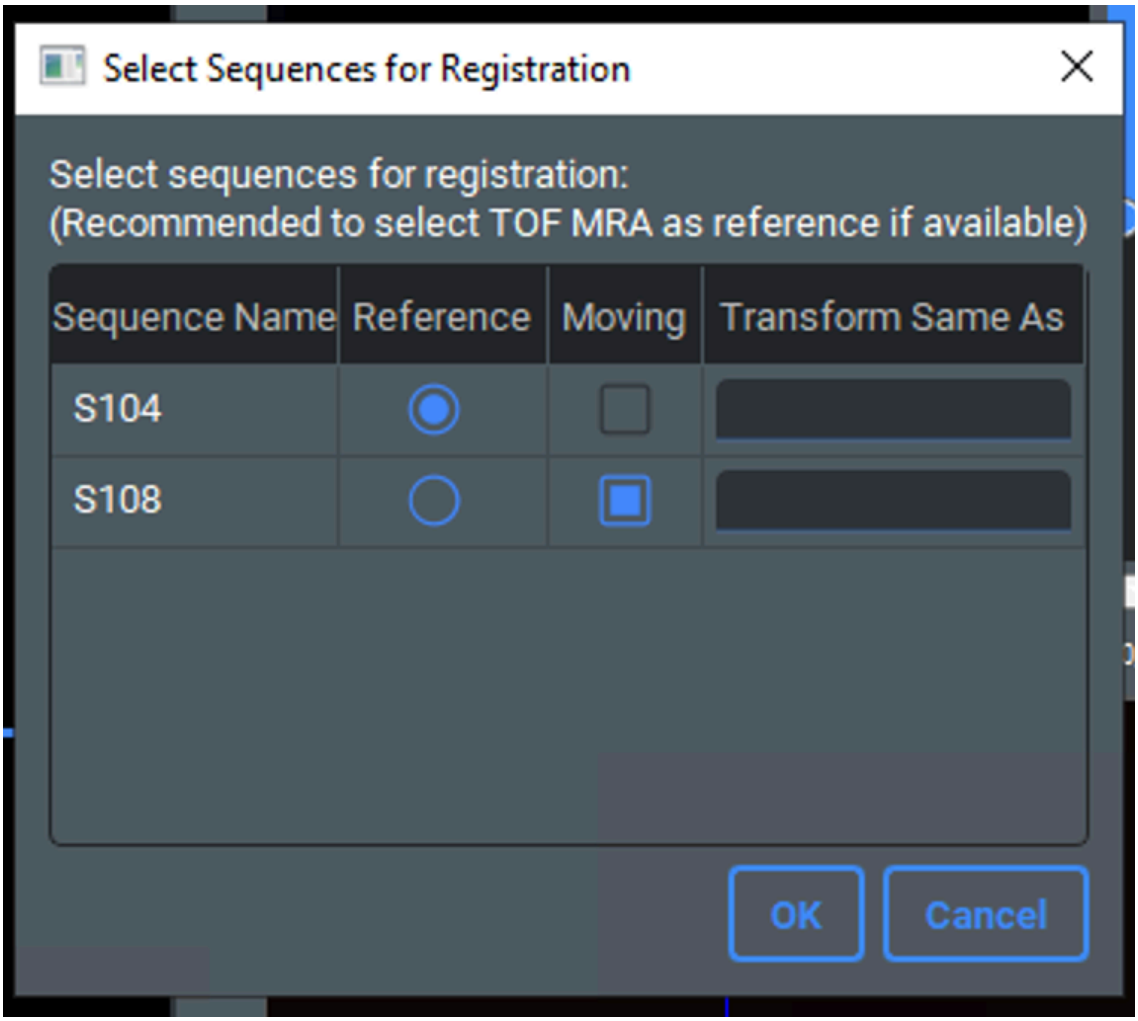
Purpose: Rigid inter-sequence alignment for `*.nii.gz` **files already in one folder** — writes `reg_<moving>_to_<reference>.txt` and updates stored volumes (same logic as after **File** → **Open DICOMs**).

Data requirement

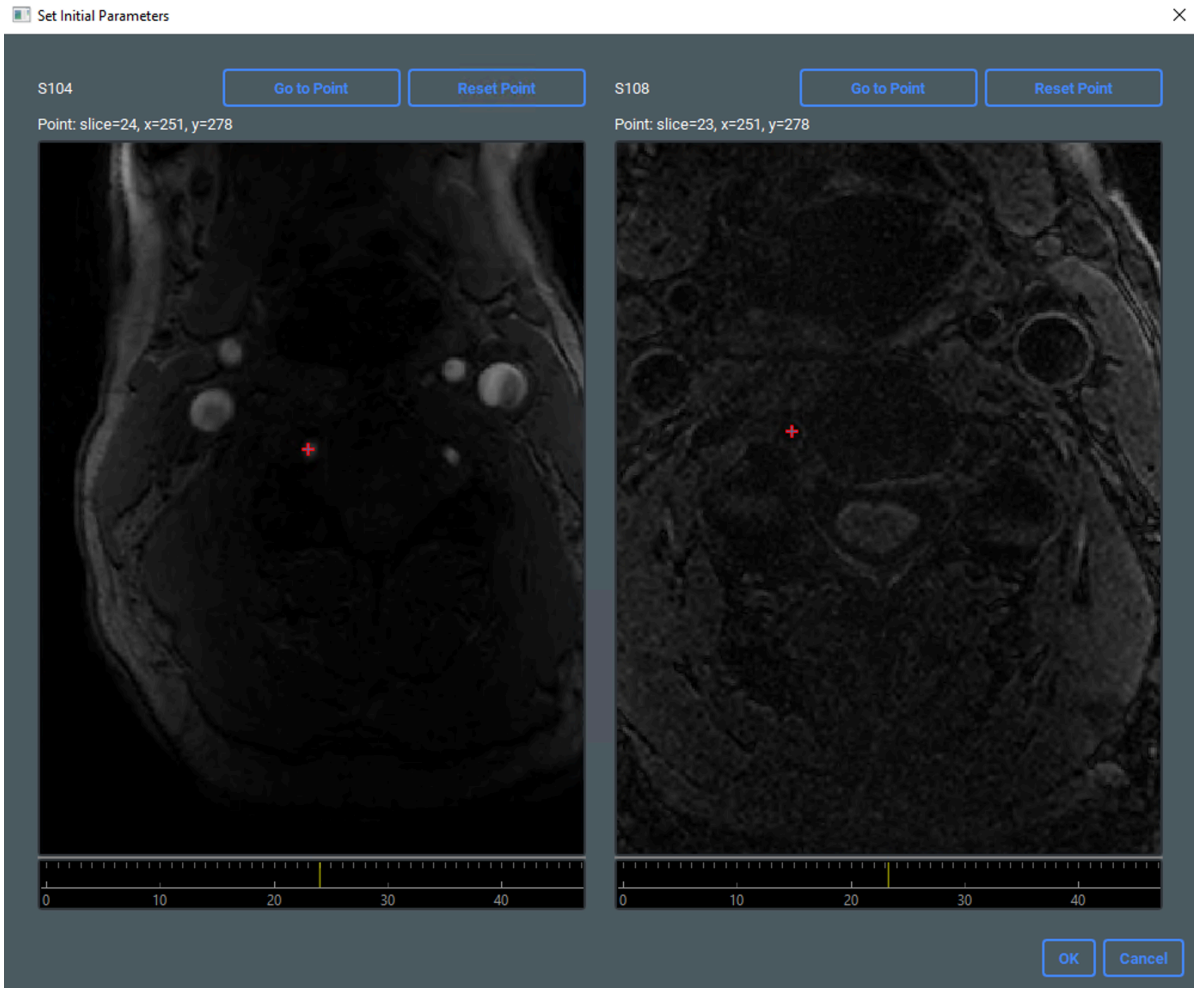
- At least one `.nii` / `.nii.gz` in the chosen folder; otherwise **No NIfTI Files** and the flow stops.

Steps

1. **Select Folder with NIfTI Files of 3D Scans** → pick the folder.
2. **Select Sequences for Registration:**
    - Caption **Select sequences for registration:** (second line recommends TOF MRA as reference when available).
    - Columns: **Sequence Name**, **Reference** (radio), **Moving** (checkboxes), **Transform Same As** (dropdown: share one moving sequence's transform).
    - **OK** / **Cancel**.



3. After **OK**: **Set Initial Parameters** (`RegistrationDialog`): per sequence — **Go to Point**, **Reset Point**, image view, label `Point: slice=..., x=..., y=...`. **Left-click** on the image moves the seed point on the current slice; **Go to Point** jumps to that slice; **Reset Point** restores the default center.



4. **OK** starts registration; **Registration Progress** may appear (metric plot + progress bar).

Outcome: `reg_<moving>_to_<reference>.txt` plus updated `{seq}.nii.gz` origins so sequences share one reference space. Opening that folder in **3D Tracing** via **File → Open Folder** and switching **Display:** keeps anatomy aligned across sequences.

1.3 Relation to File → Open DICOMs

- **File → Open DICOMs:** working directory + DICOM directory → **Select Sequences** → NIfTI + `.workflow_state.yaml` + **Workflow Banner** → same **Select Sequences for Registration** → **Set Initial Parameters** → **Registration Progress** chain; **3D Tracing** is often opened on that working directory.
- **MV: Open Raw Data:** same conversion (and optional registration) without writing workflow state.

- **Registration:** NIfTI already on disk; alignment only.

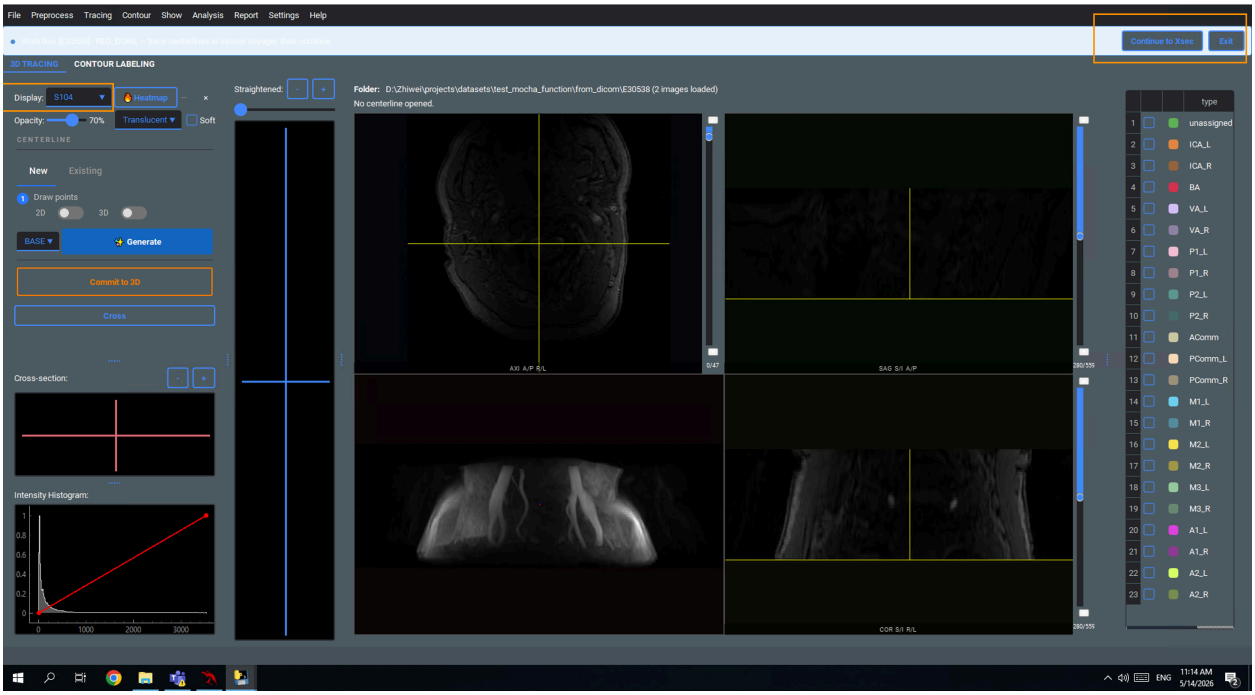
2. Three ways to load data into 3D Tracing

These paths differ only in **how** volumes (and optional centerlines) reach **3D Tracing**. **File → Resume Workflow...** restores a prior **Open DICOMs** session — not a separate data category.

#	Menu	Data requirements	After load
1	**File → Open DICOMs**	Valid DICOM; writable working directory; series selected in **Select Sequences**	NIfTI + (usually) registration; **Display:** lists sequences from the working directory
2	**File → Open Folder**	Multiple `*.nii` / `*.nii.gz` in one folder; for alignment, `reg_*_to_*.txt` from **Registration** or the workflow is recommended; optional `centerline.yaml`	**Select Images to Load** → pick loaded sequences and initial display; **Display:** switches among them
3	**File → Open**	One of: single `*.nii` / `*.nii.gz`; centerline `*.yaml` / `*.yml` / `*.swc`; single DICOM (series load); `.vts` / project launch if the build is wired for it (per file dialog filters)	Single-volume case: one **Display:** entry; yaml/swc attaches a centerline to the current volume

Tracing behavior

Once the **3D Tracing** layout is shown (left controls + central 3D + tri-planar MPR), **tracing behavior is the same**: only the number of sequences, presence of `reg_*.txt`, and optional `centerline.yaml` differ. Menu **Tracing** items **To Key Point Based**, **Add Vessel Type**, **Delete Vessel Type**, **Comparison**, and **File → Save** / **Save As...** apply to all three entry paths.



3. 3D Tracing

Switch to the **3D Tracing** tab (alongside **Contour Labeling**). Left: control column. Center: 3D volume. Around it: **Axial / Sagittal / Coronal** MPRs, each with a **vertical slider** and **arrow buttons** for slice index.

3.1 Left column (top to bottom, UI strings)

Area	Control	Role
Row 1	**Display:** + combo box	Selects the active sequence; **MPR, 3D rendering, and voxel coordinates for edits** follow this choice
Row 1 right	**Heatmap** button, **—**, **×**	Optional segmentation NIfTI overlay on 3D + MPR; **×** clears it
Row 2	**Opacity:** slider, **Translucent / Isosurface**, **Soft**	Heatmap display only; **no change to centerline data**
CENTERLINE	**New** / **Existing**	Declares whether **Commit to 3D** will **add** a segment or **replace** the selected one
Step 1 Draw points	**2D** →**Edit Sparse Points**	When on: edit **sparse MPR draft points** (not in the 3D centerline tree until **Commit to**

Area	Control	Role
		3D)
Same row	**3D → Add Points**	When on: **left-click** on the 3D volume adds points **directly** to the centerline
Step 2 Insert Points	Small toggle (no label), slot buttons, **✓ Insert**, **Clear**	Mainly under **Existing**: anchor-based single insertions on MPR; **Clear** drops anchors and anchor-inserted points
Step 3 Select	**O > 〜 / 〜 > O**	Hover priority between **key points** and **dense points** in 3D
Fit row	**BASE / VWI** + **Generate**	Auto-fit the **MPR draft**, resparse, **replace draft only** — not a save to the centerline tree
Commit	**Commit to 3D**	Writes the **MPR draft** (or internal fit candidate) into the **3D centerline**; needs **≥ 2** draft points
Bottom	**Cross**	After **exactly one** vessel is selected in 3D, jump to **Contour Labeling** for cross-sections

Menu Tracing: Add Vessel Type / **Delete Vessel Type** manage vessel labels; **To Key Point Based** downsamples dense polylines; **Comparison** opens the comparison tool. **File** → **Save** (`Ctrl+S`) / **Save As...** persist the centerline.

3.2 Two annotation modes

Mode	When to use	Where data goes
3D volume	Strong vessel contrast in the render (typical **TOF**)	**Directly** into the centerline tree; **no Commit to 3D**
2D MPR	Weak render but visible on axial/sagittal/coronal (typical **T1**, dark-blood **VWI**); or fewer manual clicks plus auto-fit	**MPR draft** first; **Commit to 3D** writes the tree

New vs Existing

- **New:** next **Commit to 3D adds** a new segment.

- **Existing:** select **one** segment in 3D, switch to **Existing** — key points load into the MPR draft; **Commit to 3D replaces** that segment's geometry.

3.3 3D volume — Add Points

1. Set **New** for a new segment (or keep an existing centerline as needed).
2. Turn **Draw points (3d)** on.
3. **Left-click** on the vessel in the central 3D view (rotate the volume as needed so the ray hits the lumen).

Effects: each valid click adds a **key point**; turning **Draw points (3d)** off ends the add session. **Ctrl+Z:** when the layout is **3D-only maximized** and **Draw points (3d)** is still on, undo typically removes the **last 3D-added point** (separate stack from MPR draft undo).

3.4 2D MPR — Edit Sparse Points

Navigation

- **Slider / arrows** beside each MPR: change slice index only.
- **Edit Sparse Points off: Left-click** on MPR usually **moves the crosshair** (browse).
- **Edit Sparse Points on: Left-click** is for **insert / select** (table below); for browse-only, turn the toggle off or use sliders.

Draft edits (Edit Sparse Points on; Existing differs for select)

Goal	Action	Result
Insert	**New: Left-click** on empty area; **Existing: Left-click** without modifiers	Insert index chosen along the polyline
Select	**New: Left-click** near an existing point; **Existing: Shift + left-click** near a point	Selection ring for delete or drag
Move	After select, **left-click** on the **same MPR and same slice**	Updates coordinates; **click on another slice clears selection** without moving

Goal	Action	Result
Delete selected	**Delete** / **Backspace**	Removes that draft point
Undo draft edit	**Ctrl+Z** / system Undo	Reverts last insert/delete/move (ordering with anchor undo as implemented)

Generate

- **Why:** Refine a sparse manual trace along the vessel.
- **How:** Choose **BASE** (bright-lumen cost) or **VWI** (dark-blood / wall-oriented cost), then **Generate** (button shows a sparkle icon + **Generate**).
- **Effect: Replaces the MPR draft** with a fitted sparse path; still **no write** to the 3D centerline until **Commit to 3D**.

Commit to 3D

- **Why:** Freeze MPR geometry into the **3D centerline** object.
- **How:** With **≥ 2** draft points, click **Commit to 3D**.
- **Effect: New** → new segment; **Existing** → replace the selected segment.

Cross

- **Why:** Open **Contour Labeling** along the selected vessel.
- **How:** Select **exactly one** segment in 3D → **Cross**.
- **Effect:** Tab switches to **Contour Labeling**; if selection is not unique, a message is printed and **no jump** occurs.



- When `Edit Sparse Points` is ON, MPR left-click is used for point editing instead of normal browsing. Use sliders/arrows for slice navigation when needed.
- `Commit to 3D` only applies to the current MPR draft. 3D `Add Points` does NOT require this step.
- `Cross` only works when exactly ONE centerline is selected in 3D.
- Make sure an existing centerline becomes highlighted after clicking it before editing.
- In `Edit` mode, clicking a point in the `Straightened View` will synchronize and display the location in the MPR view.
- For folders with multiple sequences, after finishing one vessel, it is recommended to:

  ```
  Cross → Contour Labeling → switch Display
  ```

  to avoid unfinished MPR drafts becoming inconsistent with the current volume.
- Before changing `Display`, it is recommended to finish the current MPR draft and `Commit to 3D`.
- Keyboard shortcuts:

Key	Function
G	3D View
M	MPR View

- `Open Folder` expects only one consistent `reg_*_to_*` reference space. Multiple conflicting transforms may cause loading issues.
- `.workflow_state.yaml` and Workflow Banner are only created through:

  ```
  File → Open DICOMs
  ```